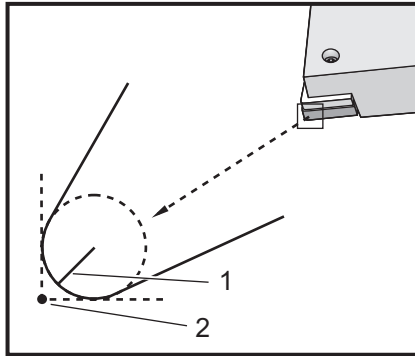


F5.10: Imaginary tool tip: [1] Tool nose radius, [2] Imaginary tool tip.



Tool nose compensation assumes that a compensated tool has a radius at the tool tip that it must compensate for. This is called the Tool Nose Radius. Since it is difficult to determine exactly where the center of this radius is, a tool is usually set up using what is called the Imaginary Tool Tip. The control also needs to know which direction the tool tip is relative to the center of the tool nose radius, or the Tip direction. The tip direction should be specified for each tool.

The first compensated move is generally a move from a non-compensated position to a compensated position and is therefore unusual. This first move is called the Approach move and is required when using tool nose compensation. Similarly, a Depart move is required. In a Depart move, the control moves from a compensated position to a non-compensated position. A Depart move occurs when tool nose compensation is canceled with a G40 command or Txx00 command. Although Approach and Depart moves can be precisely planned, they are generally uncontrolled moves and the tool should not be in contact with the part when they occur.

5.7.3 Using Tool Nose Compensation

The following steps are used to program a part using TNC:

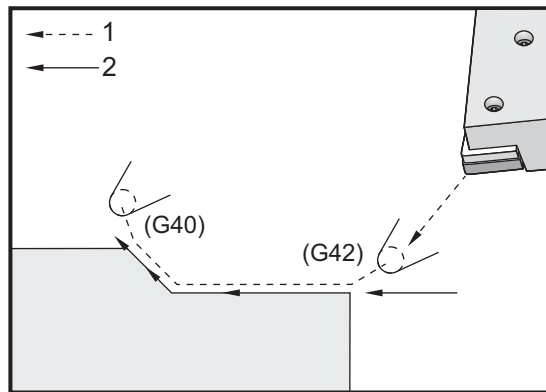
1. **Program** the part to finished dimensions.
2. **Approach and Departure** – Ensure that there is an approach move for each compensated path and determine which direction (G41 or G42) is used. Ensure that there is also a departure move for each compensated path.
3. **Tool Nose Radius and Wear** – Select a standard insert (tool with radius) to be used for each tool. Set the tool nose radius of each compensated tool. Clear the corresponding tool nose wear offset to zero for each tool.
4. **Tool Tip Direction** – Input the tool tip direction for each tool that is using compensation, G41 or G42.
5. **Tool Geometry Offset**– Set the tool length geometry and clear the length wear offsets of each tool.

6. **Check Compensation Geometry** – Debug the program in graphics mode and correct any tool nose compensation geometry problems that may occur. A problem can be detected in two ways: an alarm is generated indicating compensation interference, or the incorrect geometry is seen generated in graphics mode.
7. **Run and Inspect First Article** – Adjust compensated wear for the setup part.

5.7.4 Approach and Departure Moves For TNC

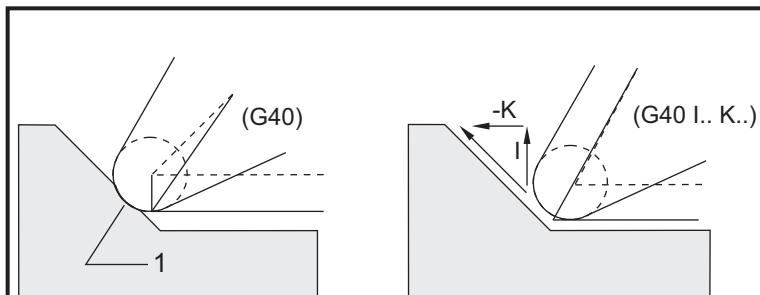
The first X or Z motion in the same line that contains a G41 or G42 is called the Approach move. The approach must be a linear move, that is a G01 or G00. The first move is not compensated, yet at the end of the approach move the machine position is fully compensated. See the following figure.

F5.11: TNC Approach and Depart Moves: [1] Compensated Path, [2] Programmed path.



Any line of code with a G40 cancels tool nose compensation and is called the Departure move. The departure must be a linear move, that is a G01 or G00. The start of a departure move is fully compensated; the position at this point is at a right angle to the last programmed block. At the end of the departure move the machine position is not compensated. See the previous figure.

The following figure shows the condition just prior to canceling tool nose compensation. Some geometries result in over or undercutting of the part. This is controlled by including an I and K address code in the G40 cancellation block. The I and K in a G40 block define a vector that is used to determine the compensated target position of the previous block. The vector is usually aligned with an edge or wall of the completed part. The following figure shows how I and K correct undesired cutting in a departure move.



5.7.5 Tool Nose Radius and Wear Offset

Each turning tool that uses tool nose compensation requires a Tool Nose Radius. The tool tip (tool nose radius) specifies how much the control is to compensate for a given tool. If standard inserts are being used for the tool, then the tool nose radius is simply the tool tip radius of the insert.

Associated with each tool on the geometry offsets page is a Tool Nose Radius Offset. The column labeled **Radius** contains the value for the tool nose radius of each tool. If the value of any tool nose radius offset is set to zero, no compensation is generated for that tool.

Associated with each radius offset is a Radius Wear Offset, located on the **Wear Offset** page. The control adds the wear offset to the radius offset to obtain an effective radius that is used for generating compensated values.

Small adjustments (positive values) to the radius offset during production runs should be placed in the wear offset page. This allows the operator to easily track the wear for a given tool. As a tool is used, the insert generally wears so that there is a larger radius at the end of the tool. When replacing a worn tool with a new one, clear the wear offset to zero.

It is important to remember that tool nose compensation values are in terms of radius rather than diameter. This is important when tool nose compensation is canceled. If the incremental distance of a compensated departure move is not twice the radius of the cutting tool, overcutting occurs. Always remember that programmed paths are in terms of diameter and allow for twice the tool radius on departure moves. The Q block of canned cycles that require a **PQ** sequence is often a departure move. The following example illustrates how incorrect programming results in overcutting.

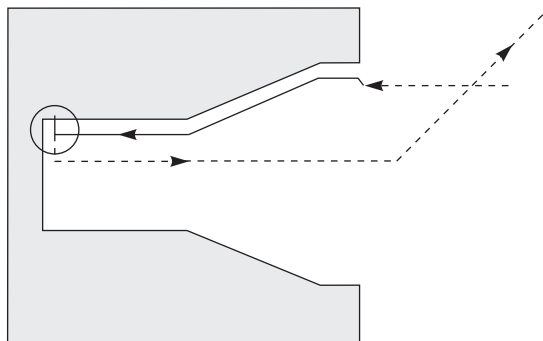
Preparation:

Tool Geometry	X	Z	Radius	Tip
8	-8.0000	-8.00000	.0160	2

Example:

```
%  
o30411 (TOOL NOSE RADIUS AND WEAR OFFSET) ;  
(G54 X0 is at the center of rotation) ;  
(Z0 is on the face of the part) ;  
(T1 is a boring bar) ;  
(BEGIN PREPARATION BLOCKS) ;  
T101 (Select tool and offset 1) ;  
G00 G18 G20 G40 G80 G99 (Safe startup) ;  
G50 S1000 (Limit spindle to 1000 RPM) ;  
G97 S500 M03 (CSS off, Spindle on CW) ;  
G00 G54 X0.49 Z0.05 (Rapid to 1st position) ;  
M08 (Coolant on) ;  
(BEGIN CUTTING BLOCKS) ;  
G96 S750 (CSS on) ;  
G41 G01 X.5156 F.004 (TNC left on) ;  
Z-.05 (Linear feed) ;  
X.3438 Z-.25 (Linear feed) ;  
Z-.5 (Linear feed) ;  
X.33 (Linear feed) ;  
G40 G00 X0.25 (TNC off, exit line) ;  
(BEGIN COMPLETION BLOCKS) ;  
G00 Z0.1 M09 (Rapid retract, coolant off) ;  
G53 X0 (X home) ;  
G53 Z0 M05 (Z home, spindle off) ;  
M30 (End program) ;  
%
```

F5.13: TNC Departure Cutting Error



5.7.6 Tool Nose Compensation and Tool Length Geometry

You set up the length geometries for tools that use tool nose compensation in the same way that you set up tools that do not use compensation.

Refer to page **94** for details on touching off tools and recording tool length geometries. When you set up a new tool, be sure to clear the geometry wear to zero.

If you command particularly heavy cuts on one edge of a tool, the tool can wear unevenly. In this case, adjust the **X or Z Geometry Wear** instead of the **Radius Wear**. You can often adjust X or Z length geometry wear to compensate for uneven tool nose wear. Length geometry wear shifts all dimensions for a single axis.

The program design may not let you use length geometry shift to compensate for wear. To determine which wear to adjust, check several X and Z dimensions on a finished part. Wear that is even results in similar dimensional changes on the X and Z axes, and suggests that you should increase the radius wear offset. Wear that affects the dimensions on one axis only suggests length geometry wear.

Good program design based on the geometry of the part should eliminate uneven wear problems. Generally, rely on finishing tools that use the entire radius of the cutter for tool nose compensation.

5.7.7 Tool Nose Compensation in Canned Cycles

Some canned cycles ignore tool nose compensation, expect a specific coding structure, or perform their own specific canned cycle activity (also refer to page **262** for more information on using canned cycles).

The following canned cycles ignore tool nose radius compensation. Cancel tool nose compensation before any of these canned cycles:

- G74 End face grooving cycle, peck drilling
- G75 O.D./I.D. grooving cycle, peck drilling
- G76 Thread cutting cycle, multiple pass
- G92 Thread cutting cycle, modal

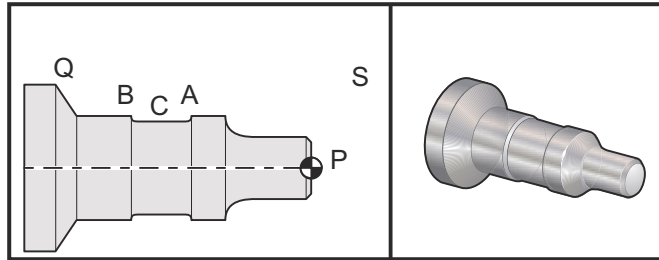
5.7.8 Example Programs Using Tool Nose Compensation

This section gives several examples of programs that use Tool Nose Compensation.

Example 1: TNC Standard Interpolation Modes G01/G02/G03

This example of general TNC uses standard interpolation modes G01/G02/G03.

F5.14: TNC Standard Interpolation G01, G02, and G03



Preparation

- Set up these tools:
 - T1 Insert with .0312 radius, roughing
 - T2 Insert with .0312 radius, finishing
 - T3 .250 wide grooving tool with .016 radius/same tool for offsets 3 and 13

Tool	Offset	X	Z	Radius	Tip
T1	01	-8.9650	-12.8470	.0312	3
T2	02	-8.9010	-12.8450	.0312	3
T3	03	-8.8400	-12.8380	.016	3
T3	13	-8.8400	-12.588	.016	4

```
O30421 (TNC STANDARD INTERPOLATION G01/G02/G03) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an rough OD tool) ;
(T2 is a finish OD tool) ;
(T3 is a groove tool) ;
(T1 PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X2.1 Z0.1 (Rapid to position S) ;
M08 (Coolant on) ;
```

```
G96 S200 (CSS on) ;
(T1 CUTTING BLOCKS) ;
G71 P1 Q2 U0.02 W0.005 D.1 F0.015 (Begin G71) ;
N1 G42 G00 X0. Z0.1 F.01 (P1 - TNC on) ;
G01 Z0 F.005 (Begin toolpath) ;
X0.65 (Linear feed) ;
X0.75 Z-0.05 (Linear feed) ;
Z-0.75 (Linear feed) ;
G02 X1.25 Z-1. R0.25 (Feed CW) ;
G01 Z-1.5 (Linear feed to position A) ;
G02 X1. Z-1.625 R0.125 (Feed CW) ;
G01 Z-2.5 (Linear feed) ;
G02 X1.25 Z-2.625 R0.125 (Feed CW to position B) ;
G01 Z-3.5 (Linear feed) ;
X2. Z-3.75 (End of toolpath) ;
N2 G00 G40 X2.1 (Q2 - TNC off) ;
(T1 COMPLETION BLOCKS) ;
G97 S500 (CSS off) ;
G53 X0 M09 (X home, coolant off) ;
G53 Z0 (Z home, clear for tool change) ;
M01 (Optional program stop) ;
(T2 PREPARATION BLOCKS) ;
T202 (T2 is a finish OD tool) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X2.1 Z0.1 (Rapid to position S) ;
M08 (Coolant on) ;
G96 S200 (CSS on) ;
(T2 CUTTING BLOCKS) ;
G70 P1 Q2 (Finish P1 - Q2 using T2, G70 and TNC) ;
(T2 COMPLETION BLOCKS) ;
G97 S500 (CSS off) ;
G53 X0 M09 (X home, coolant off) ;
G53 Z0 (Z home, clear for tool change) ;
M01 (Optional program stop) ;
(T3 PREPARATION BLOCKS) ;
T303 (T3 is a groove tool) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G54 G42 X1.5 Z-2.0 (TNC on, rapid to point C) ;
M08 (Coolant on) ;
G96 S200 (CSS on) ;
(T3 CUTTING BLOCKS) ;
G01 X1. F0.003 (Linear feed) ;
G01 Z-2.5 (Linear feed) ;
G02 X1.25 Z-2.625 R0.125 (Feed CW to position B) ;
```

```

G01 G40 X1.5 (TNC off) ;
T313 (Change offset to other side of insert) ;
G00 G41 X1.5 Z-2.125 (TNC left on) ;
G01 X1. F0.003 (Linear feed) ;
G01 Z-1.625 (Linear feed) ;
G03 X1.25 Z-1.5 R0.125 (Feed CCW to position A) ;
(T3 COMPLETION BLOCKS) ;
G00 G40 X1.6 M09 (TNC off, coolant off) ;
G97 S500 (CSS off) ;
G53 X0 (X home) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 ;

```



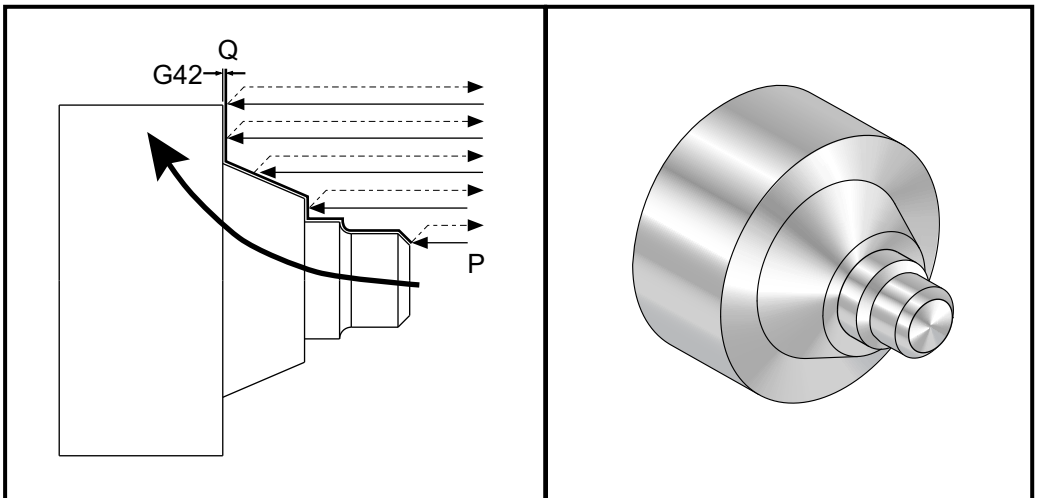
NOTE:

The suggested template of the previous section for G70 is used. Also note that compensation is enabled in the PQ sequence but is canceled after G70 is completed.

Example 2: TNC with a G71 Roughing Canned Cycle

This example uses TNC with a G71 roughing canned cycle.

F5.15: TNC G71 Roughing Canned Cycle



Preparation:

- Tools:
 - T1 Insert with 0.032 radius, roughing

Tool	Offset	Radius	Tip
T1	01	.032	3

```

o30711 (TNC WITH A G71 ROUGHING CYCLE) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an OD cutting tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X3.0 Z0.1 (Rapid to 1st position) ;
M08 (Coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G96 S200 (CSS on) ;
G71 P1 Q2 U.01 W.005 D.08 F.012 (Begin G71) ;
N1 G42 G00 X0.6 (P1 - TNC on) ;
G01 Z0 F0.01 (Begin toolpath) ;
X0.8 Z-0.1 F0.005 (45 deg. Chamfer) ;Z-0.5 (Linear feed) ;
G02 X1.0 Z-0.6 I0.1 (Feed CW) ;
G01 Z-0.9 (Linear feed) ;
X1.4 (Linear feed) ;
X2.0 Z-1.6 (23 deg. Taper) ;
G01 X3. (End of toolpath) ;
N2 G00 G40 X4. (Q2 - TNC off) ;
(BEGIN COMPLETION BLOCKS) ;
G97 S500 (CSS off) ;
G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;

```



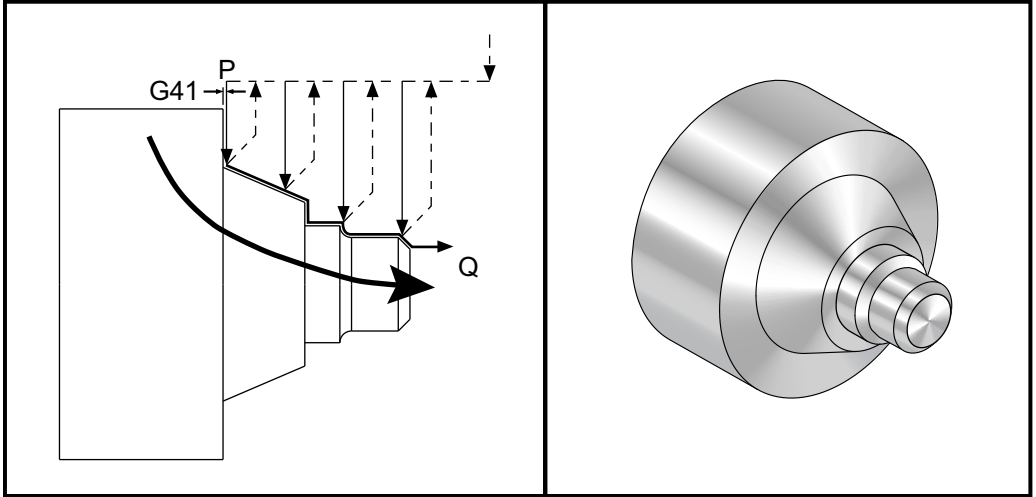
NOTE:

This part is a G71 Type I path. When using TNC it is very unusual to have a Type II path, as the compensation methods can only compensate the tool tip in one direction.

Example 3: TNC with a G72 Roughing Canned Cycle

This example is TNC with a *G72* roughing canned cycle. *G72* is used instead of *G71* because the roughing strokes in *X* are longer than the *Z* roughing strokes of a *G71*. It is therefore more efficient to use *G72*.

F5.16: TNC *G72* Roughing Canned Cycle



```
o30721 (TNC WITH A G72 ROUGHING CYCLE) ;  
(G54 X0 is at the center of rotation) ;  
(Z0 is on the face of the part) ;  
(T1 is an OD cutting tool) ;  
(BEGIN PREPARATION BLOCKS) ;  
T101 (Select tool and offset 1) ;  
G00 G18 G20 G40 G80 G99 (Safe startup) ;  
G50 S1000 (Limit spindle to 1000 RPM) ;  
G97 S500 M03 (CSS off, Spindle on CW) ;  
G00 G54 X3.1 Z0 (Rapid to 1st position) ;  
M08 (Coolant on) ;  
(BEGIN CUTTING BLOCKS) ;  
G96 S200 (CSS on) ;  
G72 P1 Q2 U.01 W.005 D.08 F.012 (Begin G72) ;  
N1 G41 G00 Z-1.6 (P1 - TNC on) ;  
G01 X2. F0.01 (Begin toolpath) ;  
X1.4 Z-0.9 (Taper) ;  
X1. (Linear feed) ;  
Z-0.6 (Linear feed) ;  
G03 X0.8 Z-0.5 R0.1 (Feed CCW) ;  
G01 Z-0.1 (Linear feed) ;  
X0.7 Z0 (Chamfer, End of toolpath) ;  
N2 G00 G40 Z0.1 (Q2 - TNC off) ;
```

```

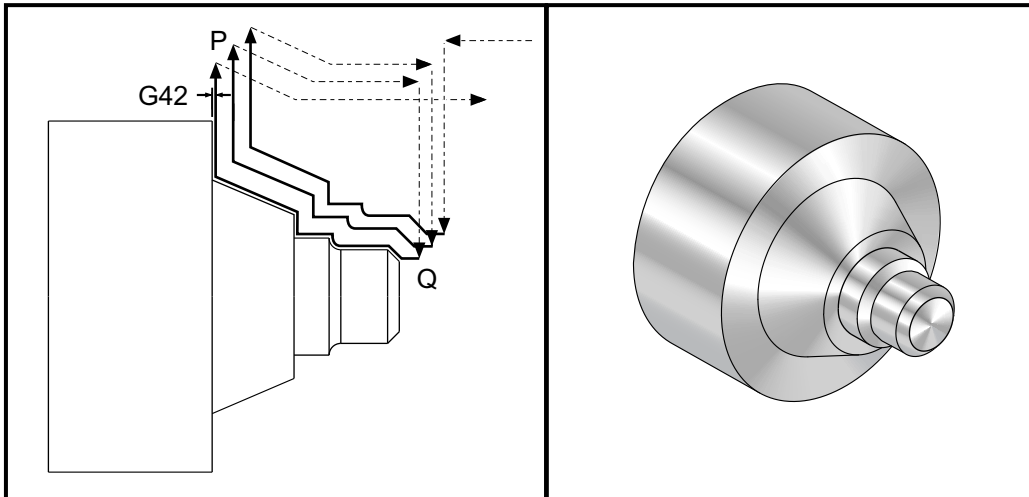
(BEGIN COMPLETION BLOCKS) ;
G97 S500 (CSS off) ;
G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;

```

Example 4: TNC with G73 Roughing Canned Cycle

This example is TNC with a G73 roughing canned cycle. G73 is best used when you want to remove a consistent amount of material in both the X and Z axes.

F5.17: TNC G73 Roughing Canned Cycle



```

O30731 (TNC WITH A G73 ROUGHING CYCLE) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on the face of the part) ;
(T1 is an OD cutting tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X3.0 Z0.1 (Rapid to 1st position) ;
M08 (Coolant on) ;
(BEGIN CUTTING BLOCKS) ;
G96 S200 (CSS on) ;
G73 P1 Q2 U.01 W.005 I0.3 K0.15 D3 F.012 (Begin G73) ;
N1 G42 G00 X0.6 (P1- TNC on) ;
G01 Z0 F0.01 (Begin toolpath) ;
X0.8 Z-0.1 F0.005 (Chamfer) ;

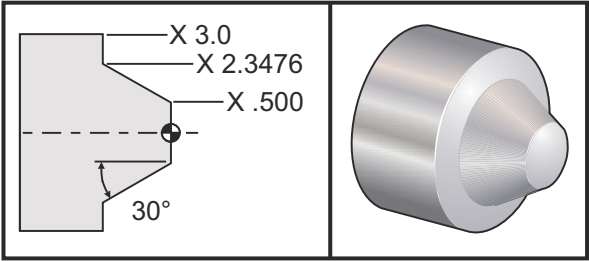
```

```
Z-0.5 (Linear feed) ;
G02 X1.0 Z-0.6 I0.1 (Feed CW) ;
G01 Z-0.9 (Linear feed) ;
X1.4 (Linear feed) ;
X2.0 Z-1.6 (Taper) ;
G01 X3. (End of toolpath) ;
N2 G00 G40 X4. (Q2 - TNC off) ;
(BEGIN COMPLETION BLOCKS) ;
G97 S500 (CSS off) ;
G53 X0 M09 (X home, coolant off) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
```

Example 5: TNC with G90 Modal Rough Turning Cycle

This example is TNC with a G90 modal rough turning cycle.

F5.18: TNC With G90 Rough Turning Cycle



Operation	Tool	Offset	Tool Nose Radius	Tip
roughing	T1	01	0.032	3

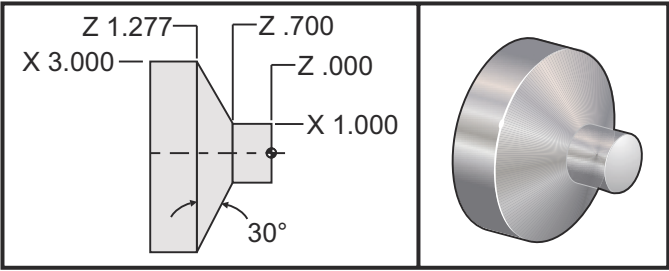
```
o30901 (TNC WITH A G90 ROUGHING CYCLE) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on face of the part) ;
(T1 is an OD cutting tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X4.0 Z0.1 (Rapid to 1st position) ;
```

```
M08 (Coolant on) ;
G96 S200 (CSS on) ;
(BEGIN CUTTING BLOCKS);
G90 G42 X2.55 Z-1.5 I-0.9238 F0.012 (Begin G90) ;
X2.45 (Optional additional pass) ;
X2.3476 (Optional additional pass) ;
(BEGIN COMPLETION BLOCKS) ;
G00 G40 X3.0 Z0.1 M09 (TNC off, coolant off) ;
G97 S500 (CSS off) ;
G53 X0 (X home) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;
```

Example 6: TNC with G94 Modal Rough Turning Cycle

This example is TNC with a G94 modal rough turning cycle.

F5.19: TNC G94 Rough Turning Cycle



Operation	Tool	Offset	Tool Nose Radius	Tip
roughing	T1	01	0.032	3

```
o30941 (TNC WITH G94 MODAL TURNING CYCLE) ;
(G54 X0 is at the center of rotation) ;
(Z0 is on face of the part) ;
(T1 is an OD cutting tool) ;
(BEGIN PREPARATION BLOCKS) ;
T101 (Select tool and offset 1) ;
G00 G18 G20 G40 G80 G99 (Safe startup) ;
G50 S1000 (Limit spindle to 1000 RPM) ;
G97 S500 M03 (CSS off, Spindle on CW) ;
G00 G54 X3.1 Z0.1 (Rapid to 1st position) ;
```

```

M08 (Coolant on) ;
G96 S200 (CSS on) ;
(BEGIN CUTTING BLOCKS) ;
G94 G41 X1.0 Z-0.5 K-0.577 F.03 (Begin G94 w/ TNC) ;
Z-0.6 (Optional additional pass) ;
Z-0.7 (Optional additional pass) ;
(BEGIN COMPLETION BLOCKS) ;
G00 G40 X3.1 Z0.1 M09 (TNC off, coolant off) ;
G97 S500 (CSS off) ;
G53 X0 (X home) ;
G53 Z0 M05 (Z home, spindle off) ;
M30 (End program) ;

```

5.7.9 Imaginary Tool Tip and Direction

It is not easy to determine the center of a tool radius on a lathe. The cutting edges are set when a tool is touched off to record tool geometry. The control calculates where the center of the tool radius is by using the edge information, the tool radius, and the direction the cutter is expected to cut in. The X- and Z-axis geometry offsets intersect at a point, called the Imaginary Tool Tip, that aids in determining the tool tip direction. The Tool Tip Direction is determined by a vector originating from the center of the tool radius and extending to the imaginary tool tip, see the following figures.

The tool tip direction of each tool is coded as a single integer from 0 to 9. The tip direction code is found next to the radius offset on the geometry offsets page. It is recommended that a tip direction be specified for all tools using tool nose compensation. The following figure is a summary of the tip coding scheme along with cutter orientation examples.



NOTE:

The tip indicates to the setup person how the programmer intends to measure the tool offset geometry. For example, if the setup sheet shows tip direction 8, the programmer intends the tool geometry to be at the edge of and on the centerline of the tool insert.